

ALGEBRA 2 - 6.3 - BINOMIAL RADICAL EXPRESSIONS

What was the rule for being able to multiply or divide radicals?

They must have the SAME INDEX.

I. ADDING AND SUBTRACTING RADICALS

We can add and subtract radicals as long as they have the SAME INDEX AND the SAME RADICAND. In a sense, we call these like radicals as opposed to like terms.

Examples. Express in simplest form.

1. $\sqrt{6} + 3\sqrt{6}$
 $4\sqrt{6}$

2. $5\sqrt[3]{5} - 2\sqrt[3]{5}$
 $3\sqrt[3]{5}$

3. $\sqrt{7} + \sqrt[3]{7}$
 can't do -
 Not like

Now, like radicals will not always be readily apparent. Sometimes we have to simplify first.

Examples. Express in simplest form.

4. $\sqrt{50} + \sqrt{18}$
 $5\sqrt{2} + 3\sqrt{2}$ (like)
 $8\sqrt{2}$

5. $7\sqrt[3]{16} - 2\sqrt[3]{54}$
 $14\sqrt[3]{2} - 6\sqrt[3]{2}$
 $8\sqrt[3]{2}$

6. $\sqrt{27} + \sqrt{45} - \sqrt{80}$
 $3\sqrt{3} + 3\sqrt{5} - 4\sqrt{5}$
 $3\sqrt{3} - \sqrt{5}$

II. MULTIPLYING BINOMIALS WITH RADICALS

Quick trip back to Algebra 1....multiply $(2x - 1)(3x + 4)$:

Guess what...that is exactly how we multiply the following...

$(2 + 3\sqrt{2})(5 - 2\sqrt{2})$
 $10 - 4\sqrt{2} + 15\sqrt{2} - 6 \cdot 2$
 $10 + 11\sqrt{2} - 12$
 $-2 + 11\sqrt{2}$

$(1 + 7\sqrt{3})(1 - 7\sqrt{3})$
 $1 - 7\sqrt{3} + 7\sqrt{3} - 49 \cdot 3$
 $1 - 49 \cdot 3$
 $1 - 147$
 -146

$(5 + 2\sqrt{7})^2$ FOIL
 $(5 + 2\sqrt{7})(5 + 2\sqrt{7})$
 $25 + 10\sqrt{7} + 10\sqrt{7} + 4 \cdot 7$
 $25 + 20\sqrt{7} + 28$
 $53 + 20\sqrt{7}$

III. DIVIDING BINOMIALS WITH RADICALS (aka Rationalizing)

How do we rationalize fractions with radicals? Multiply by an equivalent Form of 1.

In other words, you must multiply BOTH the top and bottom by the same thing to eliminate the radical.

If the denominator is a BINOMIAL (2-terms), multiply by the conjugate

Examples of conjugates: $3 - \sqrt{13} \rightarrow 3 + \sqrt{13}$

$-10 + \sqrt{11} \rightarrow -10 - \sqrt{11}$

only changes middle!

Examples. Express in simplest form.

1. $\frac{\frac{2}{\sqrt{5}} \sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$

2. $\frac{\frac{5}{3\sqrt{7}} \sqrt{7}}{\sqrt{7}} = \frac{5\sqrt{7}}{3 \cdot 7}$
 $= \boxed{\frac{5\sqrt{7}}{21}}$

3. $\frac{5}{(2+\sqrt{3})} \cdot \frac{(2-\sqrt{3})}{(2-\sqrt{3})}$ FOIL!

$$\frac{10 - 5\sqrt{3}}{4 - 2\sqrt{3} + 2\sqrt{3} - \cancel{3 \cdot 3}}$$

$$\frac{10 - 5\sqrt{3}}{4 - 3} = \boxed{\frac{10 - 5\sqrt{3}}{1}}$$

4. $\frac{\sqrt{2}}{4 - 2\sqrt{2}}$

5. $\frac{(1+\sqrt{5})}{(2-\sqrt{5})} \cdot \frac{(2+\sqrt{5})}{(2+\sqrt{5})} = \frac{2 + \sqrt{5} + 2\sqrt{5} + \cancel{5 \cdot 5}}{4 + \cancel{2\sqrt{5}} - \cancel{2\sqrt{5}} - \sqrt{5 \cdot 5}} = \frac{2 + 3\sqrt{5} + 5}{4 - 5}$

$$= \frac{7 + 3\sqrt{5}}{-1}$$

or $\boxed{-7 - 3\sqrt{5}}$